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Aim of the experiment :-

Design, assemble and test combinational circuit for adders and subtractors.

Objectives :-

1. To design and implement half adder, full adder, full subtractor and full subtractor circuits
2. To verify the outputs using their respective truth Table

Apparatus Required :-

Sl. NO	Name of the Component	Specification	Quantity
1.	Trainer Kit Analog and Digital	XPO - ANADIG1	1
2.	Bread board	—	1
3	XOR Gate (IC :- 74LS86)	Quad - 2input XOR	1
4	AND Gate (IC-74LS08)	Quad-2input AND	1
5.	OR Gate (IC-74LS32)	Quad-2input OR	1
6	NOT Gate (IC-74LS04)	Hex- inverter	1
7	Connecting wires	—	As per required

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Half Adder :-

Boolean Expression :-

$$\text{Sum}(S) = \bar{A}B + A\bar{B} = A \oplus B$$

$$\text{Carry out}(C_o) = AB$$

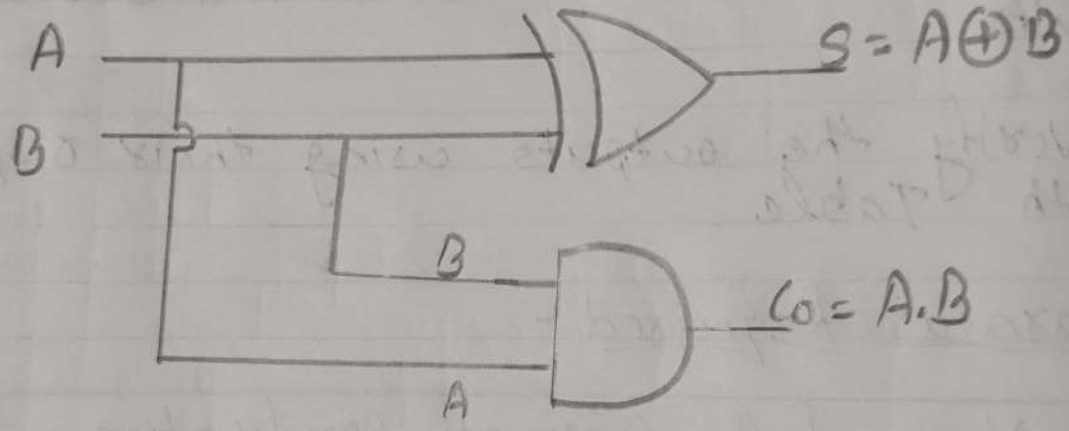


Fig 4.1 : Logic circuit of Half Adder

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Theory :-

A combinational circuit is a digital logic circuit in which the output depends only on the present values of input variables. Adders and subtractors are important example of combinational circuit used in arithmetic operation

Half Adder :-

A half adder is a combinational circuit used to add two 1 bit binary numbers It has two inputs A and B and two output sum (S) and carry (C_o).

The boolean expressions are :-

Sum (S) = $A \oplus B$

Carry (C_o) = $A \cdot B$

Thus, a half adder can be constructed using an XOR gate for sum and AND gate for carry

Truth Table for half Adder :-

Table 4.1 :-

input		output	
A	B	S	C_o
0	0	0	0
0	1	1	0
1	0	1	0
1	1	1	1

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Full Adder :-

Boolean Expression:

Sum (S) = $A \oplus B \oplus C$

Carry out (C_o) = $AB + BC + CA$

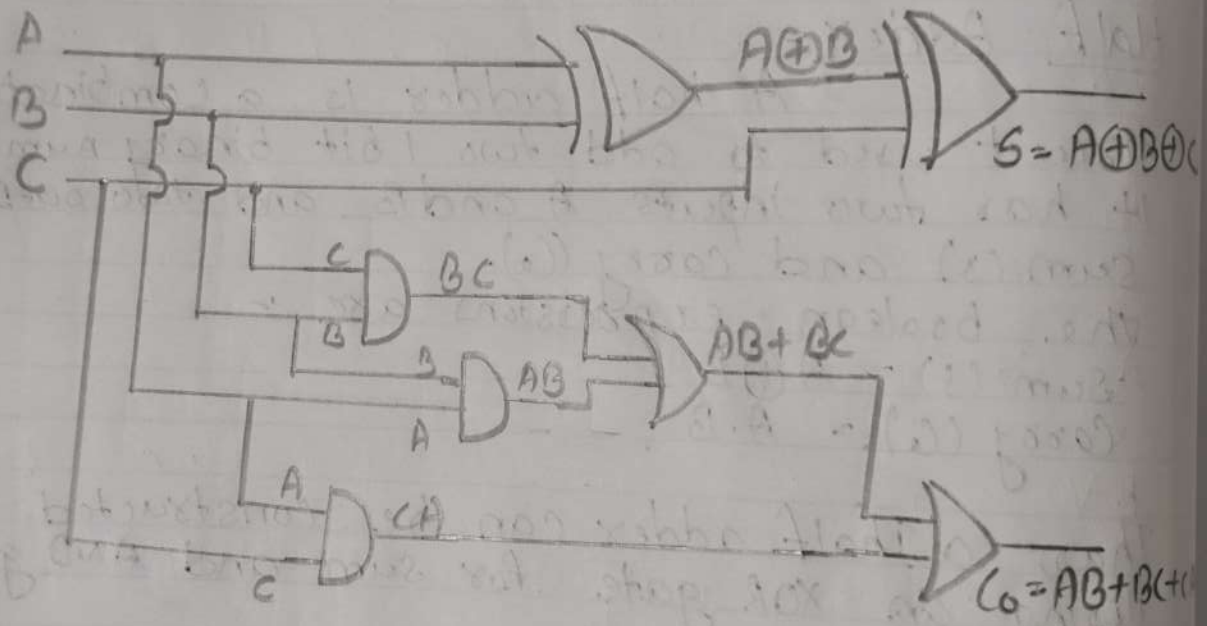


Fig 4.2:- Logic circuit Diagram of full Adder

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Full Adder

A full adder is a combinational circuit that adds three 1-bit binary input A, B, and C. It produces two output, sum(s) and carry out (Co).

Boolean expression :-

Sum (s) = $A \oplus B \oplus C$

Carry (Co) = $AB + BC + CA$

A full adder can be implemented using two XOR gate for Sum (s) and three AND gates and 2 OR gates for Carry out

Truth Table for full adder :-

Table 4.2 :-

Input			output	
A	B	C	S	Co
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

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Half Subtractor :-

Boolean expression :-

Difference (D) = $A \oplus B$

Borrow (B_o) = $\bar{A} B$

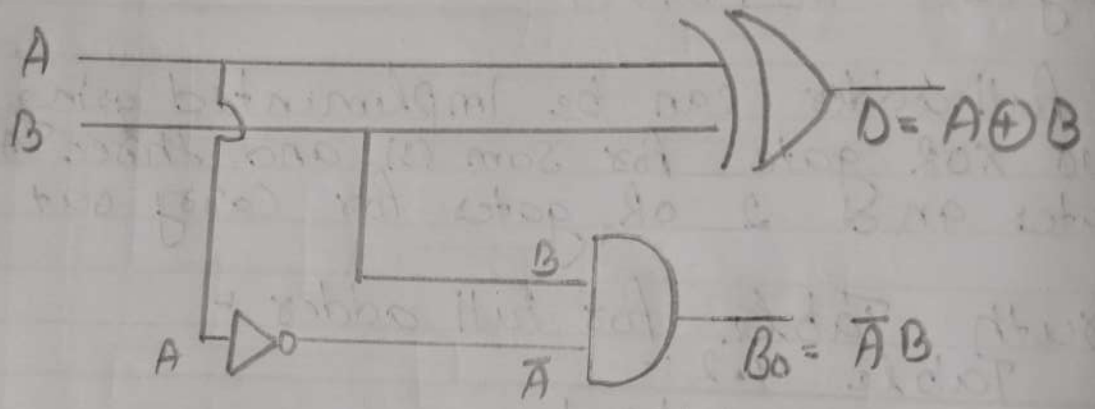


Fig 4.3:- Logic circuit Diagram of Half Subtractor

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Half Subtractor

A half subtractor is a combinational circuit used to subtract one 1-bit binary number from another. It has two input A and B and two outputs, Difference (D) and Borrow (B_o)

Boolean expression :-

$D = A \oplus B$

$B_o = \bar{A}B$

It can be constructed by XOR gate for Difference, Not gate and AND gate for Borrow.

Truth Table for half Subtractor :-

Table 4.3 :-

Input		output	
A	B	D	B_o
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

Full subtractor

A full subtractor is a Combinati circuit that perform subtraction of three 1-bit inputs : A, B and C. It produce Difference (D) and Borrow (B_o)

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Full Subtractor :-

Boolean expression :-

Difference (D) = $A \oplus B \oplus C$

Borrow (B_0) = $\bar{A}B + BC + \bar{A}C$

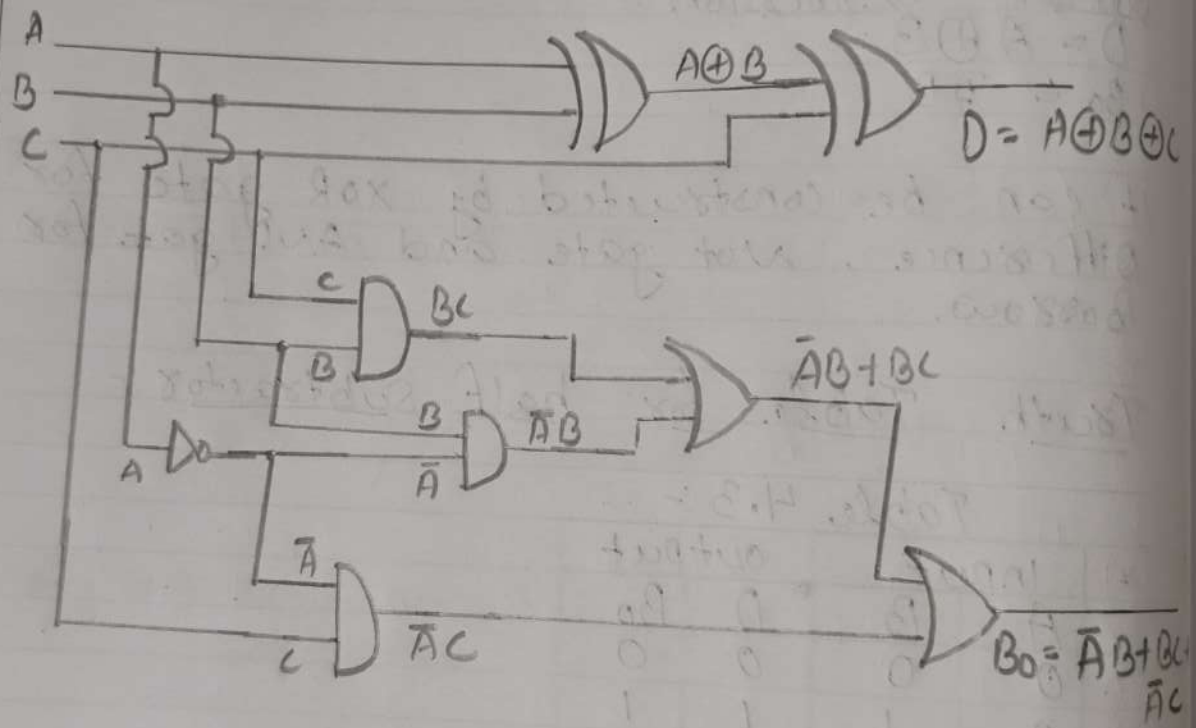


Fig 4.4 :- Logic Circuit Diagram of Full Subtractor

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Boolean expression :-

$D = A \oplus B \oplus C$

$B_o = \bar{A}B + BC + \bar{A}C$

A full subtractor can be implemented using two XOR gates and one Not and three AND, 2 OR gates.

Truth Table for full subtractor

Table 4.4:-

Input			Output	
A	B	C	D	B_o
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

Procedure :-

1. The required logic gates and digital ICs were selected for implementing the half adder, full adder, half subtractor and full subtractor circuits.

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2. The ICs were connected properly on the digital electronic trainer kit accordingly to the respective circuit diagrams.

3. The required power supply was provided to the ICs and the connections were verified

4. Different combination of binary inputs were applied to the circuits using the input switches.

5. The corresponding Sum, carry, Difference and Borrow outputs were observed and recorded using the indicator LEDs.

6. The testing was repeated for all possible combinations of input signals.

7. The observed outputs were compared with the respective theoretical truth tables and the operation of the circuit was verified.

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Fig 7.5

Fig 4.6

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OF

(ON=

2. Obse

A (O=
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OFF
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ON
ON
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(ON = 1

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Observation Table :-

1. Observation table for half Adder

Observation Table 1 :-

Input		Output		Remarks
A (D_7)	B (D_6)	S (6)	C_o (7)	
OFF	OFF	NG	NG	
OFF	ON	G	NG	The practical output matches with theoritical output
ON	OFF	G	NG	
ON	ON	G	G	

(ON = 1, OFF = 0) (NG = Not Glowing, G = Glowing)

2. Observation table for Full adder

Observation Table 2 :-

Input			Output		Remarks
A (D_7)	B (D_6)	C (D_5)	S (7)	C_o (6)	
OFF	OFF	OFF	NG	NG	
OFF	OFF	ON	G	NG	
OFF	ON	OFF	G	NG	The practical output matches with theoritical output
OFF	ON	ON	NG	G	
ON	OFF	OFF	G	NG	
ON	OFF	ON	NG	G	
ON	ON	OFF	NG	G	
ON	ON	ON	G	G	

(ON = 1, OFF = 0) (NG = Not Glowing, G = Glowing)

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Fig 4.7

Fig 4.8

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3. Observation Table for half subtractor

Observation Table 3:-

Input		Output		Remarks
A(D_7)	B(D_6)	D(6)	B_0(7)	
OFF	OFF	NG	NG	
OFF	ON	G	G	The pratical output matches with theoretical output
ON	OFF	G	NG	
ON	ON	NG	NG	

(ON=1, OFF=0) (NG=Not Glowing, G=Glowing)

4. Observation Table for full subtractor

Observation Table 4:-

Input			Output		Remarks
A(D_7)	B(D_6)	C(D_5)	D(7)	B_0(6)	
OFF	OFF	OFF	NG	NG	
OFF	OFF	ON	G	G	The practical Output matches with theoritical output
OFF	ON	OFF	G	G	
OFF	ON	ON	NG	G	
ON	OFF	OFF	G	NG	
ON	OFF	ON	NG	NG	
ON	ON	OFF	NG	NG	
ON	ON	ON	G	G	

(ON=1, OFF=0) (NG=Not Glowing, G=Glowing)

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Conclusion

Hence, the half adder, full adder half subtractor and full subtractor circuit have been successfully designed, assembled, and tested using the required logic gates. The obtained outputs have been verified against the corresponding truth table., and the desired operation of the combinational circuits has been confirmed, thereby achieving the objective of the experiment.

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